

Quick Recall: Tree characterizations

Tree (Def. 5.1.1). A graph G is *acyclic* if it contains no cycle. A *tree* is a connected acyclic graph. A vertex of degree 1 is a *leaf*.

End-vertex Lemma (5.1.3). Every tree on ≥ 2 vertices has at least 2 leaves.

Tree-growing Lemma (5.1.4). If v is a leaf of G , then G is a tree iff $G - v$ is a tree.

Theorem 5.1.2. The following are equivalent for a graph $G = (V, E)$:

- (i) G is a tree.
- (ii) (*Path uniqueness*) Between every $x, y \in V$ there is exactly one path.
- (iii) G is connected, and removing any edge disconnects G .
- (iv) G is acyclic, and adding any non-edge creates a cycle.
- (v) G is connected and $|V| = |E| + 1$.

Practice: Textbook Exercise 5.1.3* — a path-uniqueness characterization

Let $n \geq 3$ and let $G = (V, E)$ be a graph on n vertices. Prove that the following are equivalent:

- (I) G is a tree.
- (II) $G \not\cong K_n$, and for every non-edge $\{x, y\} \in \binom{V}{2} \setminus E$ there is exactly one path in G from x to y .
 - (a) [Easier direction] Prove (I) $\Rightarrow$ (II). *Hint: Why is K_n not a tree for $n \geq 3$? After ruling that out, which condition in Theorem 5.1.2 does (II) follow from?*
 - (b) [Harder direction] Prove (II) $\Rightarrow$ (I).

Hints

For (b), break the proof into two steps.

Step 1: connected. Adjacent pairs use the edge; non-adjacent pairs use (II). (*One line.*)

Step 2: acyclic. *By contradiction.* Assume G has a cycle; let $C = v_0 v_1 \cdots v_{k-1} v_0$ be a *shortest* cycle.

- **Case $k \geq 4$.** First show C has no chord (a chord would split C into two strictly shorter cycles). Then $\{v_0, v_2\}$ is a non-edge, and the two arcs of C give two distinct paths from v_0 to v_2 .
- **Case $k = 3$.** The triangle's three pairs are all edges, so we must look *outside* it.

Key trick #1: extend the triangle to a maximal clique H . (Maximality replaces an induction: *any* $w \notin H$ is forced to have a non-neighbor in H .) Some $w \notin H$ exists (else $G = K_n$). Two scenarios share the same shape—in both, find an “entry vertex” $h_0 \in H$, then use the clique to detour:

- (A) w has a neighbor $h_0 \in H$. By maximality, w has a non-neighbor $h \in H$ ($h \neq h_0$). For any $h_1 \in H \setminus \{h_0, h\}$, two paths from w to h :

$$w \rightarrow h_0 \rightarrow h \text{ (length 2)} \quad \text{vs.} \quad w \rightarrow h_0 \rightarrow h_1 \rightarrow h \text{ (length 3)}.$$

- (B) w has *no* neighbor in H . Take a shortest path $Q: w \rightsquigarrow h_0 \in H$.

Key trick #2 — switch target. Pick *any* $h \in H \setminus \{h_0\}$; then $\{w, h\}$ is automatically a non-edge. For any $h_1 \in H \setminus \{h_0, h\}$, two paths from w to h :

$$w \rightsquigarrow h_0 \rightarrow h \text{ (length } m+1) \quad \text{vs.} \quad w \rightsquigarrow h_0 \rightarrow h_1 \rightarrow h \text{ (length } m+2).$$

Answer: Part (a).

- $K_n \supseteq K_3$ contains a triangle (a cycle), so K_n is not a tree $\Rightarrow G \not\cong K_n$. ✓
- Thm 5.1.2 (i) $\Rightarrow$ (ii) gives unique paths between *all* pairs; restricting to non-edges yields (II). ✓

Part (b). Assume (II). Show G is a tree.

Step 1 (connected). Any pair: edge is a path, or (II) gives one. ✓

Step 2 (acyclic). *By contradiction.* Take a **SHORTEST** cycle $C = v_0 v_1 \cdots v_{k-1} v_0$ in G , length $k \geq 3$.

Case $k \geq 4$. *Plan: shortest $\Rightarrow$ chord-free $\Rightarrow$ a non-edge on C with two paths around it.*

- C chord-free (a chord would split C into 2 shorter cycles, $\Rightarrow \Leftarrow$).

- $\Rightarrow \{v_0, v_2\} \notin E$, and the two arcs of C give:

$$\begin{aligned} P_1: v_0 \rightarrow v_1 \rightarrow v_2 & \quad (\text{len } 2) \\ P_2: v_0 \rightarrow v_{k-1} \rightarrow \cdots \rightarrow v_2 & \quad (\text{len } k-2 \geq 2) \end{aligned}$$

- $\nexists$ uniqueness in (II). $\checkmark$

Case $k = 3$. Triangle $T = \{v_0, v_1, v_2\}$ — every pair inside T is an edge, so we must look outside it.

- *Trick:* extend T to a *maximal* clique H (greedy; $|H| \geq 3$). Maximality replaces an induction.
- $H \neq V$ (else $G = K_n$ $\nexists$). Pick $w \notin H$; by maximality, w has a non-neighbor in H .

(A) w has a neighbor $h_0 \in H$.

- Pick a non-neighbor $h \in H$ (so $h \neq h_0$) and any $h_1 \in H \setminus \{h_0, h\}$.
- Two distinct w - h paths through h_0 (using clique edges):

$$\begin{aligned} P_1: w \rightarrow h_0 \rightarrow h & \quad (\text{len } 2) \\ P_2: w \rightarrow h_0 \rightarrow h_1 \rightarrow h & \quad (\text{len } 3) \end{aligned}$$

- $\nexists$ for $\{w, h\}$. $\checkmark$

(B) w has no neighbor in H . Same shape as (A), but Q replaces the edge wh_0 .

- Take a shortest path $Q: w \rightsquigarrow h_0 \in H$, with $m \geq 2$.
- *Switch target:* every $h \in H$ is automatically a non-neighbor of w , so pick any $h \neq h_0$ (no need to track the maximality witness). Pick $h_1 \in H \setminus \{h_0, h\}$.
- Two distinct w - h paths through h_0 :

$$\begin{aligned} P_1: w \rightsquigarrow h_0 \rightarrow h & \quad (\text{len } m+1) \\ P_2: w \rightsquigarrow h_0 \rightarrow h_1 \rightarrow h & \quad (\text{len } m+2) \end{aligned}$$

- $\nexists$ for $\{w, h\}$. $\checkmark$

Conclude. Step 1 $\wedge$ Step 2 $\Rightarrow G$ is connected and acyclic, hence a tree. □